

An Evolutionary Multiobjective Optimization Based Fuzzy Method for Overlapping Community Detection

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Abstract—In the last decade, the detection of overlapping communities has received increasing attention in network science. Among various clustering techniques, the fuzzy clustering has been widely adopted in overlapping community detection, since the soft assignment provided by it naturally meets the overlapping between multiple communities. The crucial step of fuzzy clustering based overlapping community detection is to find the optimal community centers, so that the overlapping communities can be obtained according to the membership degrees between nodes and community centers. In this paper, we propose an evolutionary multiobjective optimization based fuzzy method for overlapping community detection. In contrast to traditional fuzzy clustering methods, the proposed method optimizes the community centers by using a specially tailored multiobjective evolutionary algorithm. Moreover, it can also find an appropriate fuzzy threshold for each node, so that diverse overlapping community structures can be uncovered. In the experiments, we compare the proposed method with six state-of-the-art overlapping community detection approaches on synthetic and real-world networks with different scales and characteristics. The statistical results demonstrate that the proposed method can obtain the best results on most test instances.

Index Terms—Complex network, overlapping community detection, fuzzy clustering, evolutionary multiobjective optimization.

I. INTRODUCTION

WITH the development of network science, complex networks have been applied in many fields to represent various kinds of complex systems, such as biological networks [1] and social networks [2]. Community structure denotes the division of a network into some groups, where the nodes within the same group have dense connections and those in different groups have sparse connections [3]. The detection of community structure has become an important technique in many fields, such as social science, network science, biological science, and so on [4].

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Since the community detection problem was proposed by Girvan and Newman in 2002 [3], there have been many approaches for uncovering nonoverlapping communities that each node belongs to one single community [5], [6]. However, the communities in many real-world networks are often overlapped with each other [7]. For example, one person is often a member of several social groups including clubs, friends, family, and colleagues, and a scientist may be active in several areas. In 2005, Palla *et al.* [7] extended the nonoverlapping community detection approach to uncover overlapping communities, and a variety of techniques have been adopted to detect overlapping communities from then on, including clique percolation theory [7], local expansion [8], label propagation [9], link community detection [10], fuzzy clustering [11], and evolutionary computation [12].

The fuzzy clustering has become an effective technique for overlapping community detection, since the fuzzy clustering that allows a node to belong to more than one cluster naturally meets the overlapping between multiple clusters [13]. The crucial step of fuzzy clustering based overlapping community detection is to find the optimal community centers. Once the community centers are determined, the membership matrix U between nodes and community centers can be calculated, and the overlapping communities can be obtained according to U and a predefined threshold λ . Following this idea, a number of fuzzy clustering based overlapping community detection approaches have been proposed in the last decade. For instance, Zhang *et al.* [14] proposed an overlapping community detection approach based on the combination of spectral mapping, fuzzy c -means (FCM) [15], and optimization of a quality function; Nepusz *et al.* [16] regarded the fuzzy clustering based overlapping community detection as a nonlinearly constrained optimization problem and solved it by a quadratic-time algorithm; and Wang *et al.* [17] proposed a fuzzy clustering algorithm based on local random walk and a new distance measure.

Although the above approaches have successfully employed fuzzy clustering to uncover overlapping communities, the fuzzy clustering is still criticized for the high sensitivity to community center initialization, which makes these approaches easily trapped into local optimum [18]–[20]. Besides, some parameters including the number of communities k and the fuzzy threshold λ should be predefined in these approaches, which are pivotal to the clustering result but difficult to be determined in advance [11], [21]–[23]. To address these issues, this paper proposes a fuzzy method for overlapping community detection, which uses a multiobjective evolutionary algorithm (MOEA) [24] to automatically opti-

mize the community centers as well as the parameters in fuzzy clustering. The contributions of this paper are as follows:

1) : An evolutionary multiobjective optimization based fuzzy method (EMOFM) is proposed for overlapping community detection. The proposed method adopts the idea of fuzzy c -medoids [25], where each community center is a node in the network (termed central node). The central nodes are optimized by a well tailored MOEA, and the number of communities can also be automatically determined. Moreover, the MOEA is also used to optimize the fuzzy threshold of each node, which enables the proposed method to find diverse overlapping community structures. As a result, the proposed method can adapt to various networks having different overlapping degrees without any predefined parameter.

2) : To enhance the performance of the MOEA, a two-stage optimization process is considered in the proposed method: First, the central nodes are optimized by minimizing two objectives (i.e., the kernel k -means [26] and the ratio cut [27]), which can find the optimal nonoverlapping communities. Second, the fuzzy thresholds of non-central nodes are optimized by maximizing two objectives (i.e., the extended modularity [28] and the number of overlapping nodes), which can find a good membership for each non-central node and thus able to detect overlapping communities. Furthermore, two novel population initialization strategies are designed for the two stages, respectively, which are verified to be able to significantly improve the performance of the proposed method.

3) : To validate the effectiveness of the proposed method, we implement three approaches by embedding three distance measures in EMOFM, i.e., spectral clustering based Euclidean distance, random walk based distance, and diffusion kernel similarity. According to the experimental results, it is empirically verified that the three EMOFM based approaches outperform six state-of-the-art approaches for overlapping community detection.

The rest of the paper is organized as follows. We first give some preliminaries about community detection and related work in Section II. Then, we present the details of the proposed EMOFM in Section III. Afterwards, we report the empirical results to show the effectiveness of the proposed method in Section IV. Finally, conclusions are drawn in Section V.

II. PRELIMINARIES AND RELATED WORK

A. Formal Notations

To start with, the formal notations used in this paper are listed in Table I, including those for graphs, fuzzy sets, membership matrix, and MOEA.

B. The Community Detection Problem

Formally, a network can be modeled as a graph denoted by $G = (V, E)$, where $V = \{v_1, v_2, \dots, v_n\}$ is the set of nodes, $E = \{(v_i, v_j) | v_i \in V, v_j \in V, \text{ and } i \neq j\}$ is the set of edges, and $n = |V|$ is the number of nodes. A network G can also be represented by the adjacency matrix $A = (A_{ij})^{n \times n}$. In this paper, we merely consider undirected and unweighted networks, i.e., $A_{ij} = A_{ji}$ and $A_{ij} \in \{0, 1\}$. A community

TABLE I
FORMAL NOTATIONS USED IN THIS PAPER.

Notation	Description
$G(V, E)$	a graph G with node set V and edge set E
V	a node set $V = \{v_1, v_2, \dots, v_n\}$
E	a edge set $E = \{(v_i, v_j) v_i \in V, v_j \in V, \text{ and } i \neq j\}$
n, m	number of nodes in V , number of edges in E
A	adjacency matrix of G , where $A \in \{0, 1\}^{n \times n}$
d	degree of the i -th node, where $d_i = \sum_{j=1}^n A_{ij}$
C	a community set $C = \{C_1, \dots, C_k\}$, where $C_1 \cup \dots \cup C_k = V$
k	number of communities in C
$\mathbf{b}$	a binary vector $\mathbf{b} = (b_1, b_2, \dots, b_n)$, where $\mathbf{b} \in \{0, 1\}^n$
$\mathbf{r}$	a real vector $\mathbf{r} = (r_1, r_2, \dots, r_n)$, where $\mathbf{r} \in [0, 1]^n$
CN	a central node set $CN = \{CN_1, \dots, CN_k\}$
NC	a non-central node set $NC = \{NC_1, \dots, NC_{n-k}\}$
U	membership matrix between NC and CN , $U \in \mathbb{R}^{(n-k) \times k}$
U_{ij}	membership degree between NC_i and CN_j
P	a population $P = \{\mathbf{p}_1, \mathbf{p}_2, \dots\}$
$\mathbf{p}_i$	the i -th individual in P , $\mathbf{p}_i = \{b_1, \dots, b_n, r_1, \dots, r_n\}$
$\mathbf{p}_{ib}, \mathbf{p}_{ir}$	the vector $(b_1, \dots, b_n)$ and the vector $(r_1, \dots, r_n)$ in $\mathbf{p}_i$
P'	a parent population for generating offsprings
O	an offspring population
$SubP_i$	the i -th subpopulation
N_{pop}, N_{sub}	size of population P , size of each subpopulation $SubP_i$
Gen	maximum number of generations
$stage$	current stage number
FR	final result, $FR = P \cup SubP_1 \cup \dots \cup SubP_{ SubP }$

structure is defined as $C = \{C_1, C_2, \dots, C_k\}$ in G , which satisfies the following conditions:

$$V = \bigcup_{i=1}^k C_i, \text{ where } C_i \neq \emptyset, \quad (1)$$

$$C_i \neq C_j, \forall i \neq j \text{ and } i, j \in \{1, 2, \dots, k\}. \quad (2)$$

If the set of communities satisfies:

$$C_i \cap C_j = \emptyset, \forall i \neq j \text{ and } i, j \in \{1, 2, \dots, k\}, \quad (3)$$

then C is a set of nonoverlapping communities of network G . By contrast, if the set of communities satisfies:

$$C_i \cap C_j \neq \emptyset, \exists i \neq j \text{ and } i, j \in \{1, 2, \dots, k\}, \quad (4)$$

then C is a set of overlapping communities.

C. Fuzzy Clustering Based Overlapping Community Detection

Since this paper focuses on uncovering overlapping communities, some fuzzy clustering based overlapping community detection approaches are reviewed in the following.

In [14], Zhang *et al.* devised an approach based on the fuzzy concept to identify overlapping communities in complex networks. This approach integrates fuzzy clustering with a new modularity function, namely, the spectral mapping, in which an approximation mapping of nodes into Euclidean space is calculated. Then, a soft assignment is obtained by utilizing FCM based on the Euclidean distances between nodes. Finally, the communities can be determined according to a threshold λ . Note that the number of communities k should be predefined for this approach.

In [17], a distance based on local random walk was applied to overlapping community detection. Specifically, a local distance measure is designed by using the random walk based similarity indices. Then, the membership matrix U can be obtained by FCM based on the proposed distance measure.

Similar to the approach in [14], the number of communities k and a fuzzy threshold λ should be defined in advance to get the final communities.

In contrast to the approaches based on FCM, Zhang *et al.* [29] proposed a nonnegative matrix factorization (NMF) [30] technique for fuzzy community detection, which quantified an absolute membership that a node belongs to a specific community, hence overlapping communities can be uncovered by using different membership degrees.

Psorakis *et al.* [31] developed a probabilistic approach based on Bayesian NMF model to detect overlapping communities. In this approach, the feature matrix is decomposed via NMF as part of the parameter inference for a generative model to get a soft assignment. Similarly, an extra fuzzy threshold should be predefined to get the final communities.

Binesh *et al.* [32] designed a fuzzy clustering algorithm for overlapping community detection based on NMF, which does not need an extra parameter to control the fuzziness in FCM. The experimental results demonstrated that this approach can get better membership matrix than FCM based approaches with respect to two novel evaluation criteria.

As reported in [11], [22], [23], [32], most of the existing fuzzy clustering based overlapping community detection approaches require some prior knowledge. Specifically, the number of communities k should be given for all the above approaches, and a fuzzy threshold λ should be predefined to get the final communities for the approaches in [14], [17], [31]. Besides, these approaches define a single fuzzy threshold for all nodes in the network, which is not reasonable since each node has its own trait [23]. By contrast, the proposed EMOFM assigns a separate fuzzy threshold to each node, and optimizes both the fuzzy thresholds and the number of communities via an MOEA. The idea of EMOFM is presented in Section III-A.

D. Evolutionary Computation Based Fuzzy Clustering

The fuzzy clustering can be regarded as an optimization problem, where the objective is the quality of the clustering result (e.g., fuzzy J_m index [15], Xie-Beni index [33], and overlap-separation measure [34]), and the variables are the cluster centers. The original FCM optimizes the cluster centers by iteratively updating the membership matrix and the cluster centers. However, this optimization approach is based on hill climbing methods, which are sensitive to the initial cluster centers and likely to get stuck in local optimum [18], [20].

During the last two decades, evolutionary computation has gained popularity in solving various complex optimization problems [35]. It has the following advantages in comparison to the optimization approach in FCM: Firstly, it optimizes a set of individuals (each individual corresponds to a clustering result) simultaneously, which can effectively escape from local optimum. Secondly, it can optimize not only the cluster centers, but also the parameters (e.g., the number of clusters) in FCM, thus making FCM parameterless. Thirdly, it can optimize multiple objectives simultaneously (i.e., evolutionary multiobjective optimization), which can obtain better clustering result than optimizing one single objective [36]. As a result, a number of evolutionary algorithms have been adopted

in fuzzy clustering, such as genetic algorithm [18], [37], teaching-learning-based optimization [38], particle swarm optimization [19], [20], and so on [36], [39].

In [19], Wang *et al.* suggested a particle swarm optimization based FCM for overlapping community detection. Specifically, it embeds FCM in the selection operator of particle swarm optimization, and uses it to optimize the cluster centers by minimizing the fuzzy J_m index. However, the number of clusters should be given in advance.

Saha *et al.* [36] proposed a multiobjective differential evolution based fuzzy clustering technique, which optimizes the cluster centers by minimizing both the fuzzy J_m index and the Xie-Beni index. However, the number of cluster centers should also be predefined.

To automatically determine the number of cluster centers, a multiobjective genetic algorithm NSGA-II was incorporated into FCM by Wikaisuksakul [18]. This approach also optimizes the cluster centers, but each individual has a different length so that it can represent different numbers of cluster centers. As for the objectives, it minimizes both the fuzzy J_m index and the overlap-separation measure.

In contrast to the above approaches that optimize the cluster centers, a multiobjective teaching-learning-based optimization algorithm was introduced in [38] to optimize the membership matrix. This approach minimizes the fuzzy J_m index and maximizes the partition coefficient and exponential separation, as the interaction between these two objectives is important to dynamically determine the number of clusters and explore interesting areas of the search space.

Although various evolutionary algorithms have been applied to fuzzy clustering, most of them cannot be used for overlapping community detection due to the following three reasons. Firstly, the nodes in complex networks are not in Euclidean space, hence it is difficult to represent the cluster centers. Secondly, the fuzzy threshold of each node is needed to obtain the overlapping communities, which should be optimized together with the cluster centers. Thirdly, existing evolutionary algorithms are ineffective for fuzzy clustering based overlapping community detection due to the difficulty of the problem. To address these issues, we adopt the idea of fuzzy c -medoids that forces each cluster center to be an existing node, and optimize both the cluster centers and the fuzzy thresholds. We also enhance the performance of an existing evolutionary algorithm by a two-stage optimization process and two population initialization strategies. The procedure of the proposed EMOFM is elaborated in Section III-B.

III. PROPOSED FUZZY METHOD

A. Main Idea of EMOFM

In the proposed EMOFM, each cluster center is an existing node in the network $G(V, E)$ to be clustered, termed central node. To begin with, a binary vector $\mathbf{b}$ and a real vector $\mathbf{r}$ are defined as follows:

$$\begin{aligned} \mathbf{b} &= (b_1, b_2, \dots, b_n) \in \{0, 1\}^n \\ \mathbf{r} &= (r_1, r_2, \dots, r_n) \in [0, 1]^n \end{aligned} \quad (5)$$

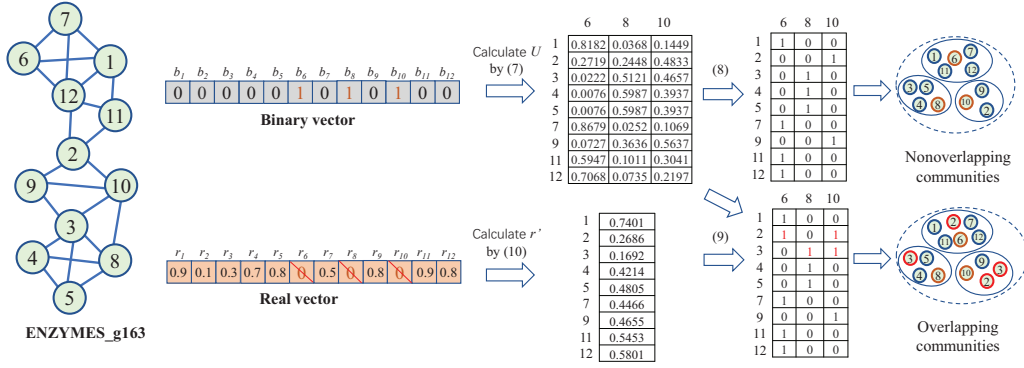


Fig. 1. An illustrative example of the proposed method to detect the communities of the ENZYMES_g163 network. The values of the binary vector and real vector are randomly generated for illustrative purposes, while they are related to the first and second optimization stages of EMOFM, respectively.

where b_i indicates whether v_i is a central node or not, r_i denotes the fuzzy threshold of v_i , and n denotes the number of nodes of the network.

To obtain the communities, the non-central node set NC and the central node set CN are first identified, i.e.,

$$\begin{aligned} NC &= \{NC_1, NC_2, \dots\} \\ CN &= \{CN_1, CN_2, \dots\}, \end{aligned} \quad (6)$$

where NC_i is the i -th node whose corresponding element in $\mathbf{b}$ is 0, and CN_i is the i -th node whose corresponding element in $\mathbf{b}$ is 1. Then, the distances between non-central nodes and central nodes are calculated based on a distance measure, and each element in the membership matrix $U = (U_{ij})^{|NC| \times |CN|}$ can be obtained by [15]

$$U_{ij} = \frac{1}{\sum_{l=1}^{|CN|} \frac{dis(NC_i, CN_j)^{\frac{2}{f-1}}}{dis(NC_i, CN_l)}}, \quad (7)$$

where U_{ij} denotes the membership degree between NC_i and CN_j , $dis(NC_i, CN_j)$ denotes the distance between the two nodes, and f is a parameter controlling the fuzziness of the resulting clusters, which was suggested to be 2 in [32]. Lastly, in order to obtain the nonoverlapping communities, each central node is regarded as the center of a community, and each non-central node NC_i belongs to the community centered at the central node CN_j if

$$U_{ij} = \max_{l=1, \dots, |CN|} U_{il}. \quad (8)$$

On the other hand, to obtain the overlapping communities, each non-central node NC_i belongs to the community centered at the central node CN_j once

$$U_{ij} \geq r'_{NC_i}, \quad (9)$$

where r'_{NC_i} is obtained by

$$r'_{NC_i} = \min_l U_{il} + r_{NC_i} \times (\max_l U_{il} - \min_l U_{il}). \quad (10)$$

As a consequence, the proposed method can detect overlapping nodes by finding proper fuzzy thresholds, where a lower value of r_{NC_i} indicates a higher probability that NC_i is an overlapping node, and vice versa. In extreme cases, $r_{NC_i} = 1$ means that NC_i belongs to a single community, and $r_{NC_i} = 0$ means that NC_i belongs to all communities.

To better understand the above procedure, Fig. 1 gives an illustrative example of the proposed method to detect the communities of the cheminformatics network ENZYMES_g163 [40], which has 12 nodes and 22 edges. Assume that the binary vector and the real vector are $\mathbf{b} = (b_1, \dots, b_{12}) = (0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0)$ and $\mathbf{r} = (r_1, \dots, r_{12}) = (0.9, 0.1, 0.3, 0.7, 0.8, 0, 0.5, 0.8, 0, 0.9, 0.8)$, respectively, where the values of $\mathbf{b}$ and $\mathbf{r}$ are randomly generated for illustrative purposes. The membership matrix U can be calculated by (7), and r' can be calculated by (10). Afterwards, the nonoverlapping communities and the overlapping communities can be obtained by (8) and (9), respectively. As shown in Fig. 1, the nonoverlapping communities of the network are $\{\{3, 4, 5, 8\}, \{1, 6, 7, 11, 12\}, \{2, 9, 10\}\}$, while the overlapping communities are $\{\{3, 4, 5, 8\}, \{1, 2, 6, 7, 11, 12\}, \{2, 3, 9, 10\}\}$ with nodes 2 and 3 being overlapping nodes. To sum up, the idea of the proposed method is entirely based on fuzzy clustering. On one hand, it calculates the membership matrix U between non-central nodes and central nodes, where U represents the membership degrees between nodes and cluster centers. On the other hand, it assigns a fuzzy threshold to each node, thus controlling whether the node is an overlapping node or not.

In the following, three methods adopted by EMOFM to measure the distances between nodes are introduced.

1) *Spectral clustering based Euclidean distance:* Spectral clustering algorithm is a clustering method based on globally optimizing cost functions [41]. In [42], the spectral clustering algorithm was first adopted in the detection of nonoverlapping communities. Later, it was adopted to detect overlapping communities by replacing k -means in spectral clustering with FCM [14]. To calculate the spectral clustering based Euclidean distance, the adjacent matrix A and the degree matrix D of the network should be obtained, where D is a diagonal matrix with each element of the principal diagonal being the degree of a node. Then, the Laplace matrix can be constructed by

$$L = D - A, \quad (11)$$

and the eigenvectors corresponding to the top $k-1$ minimum eigenvalues of L are calculated. Afterwards, a new matrix $Eig \in \mathbb{R}^{n \times (k-1)}$ is obtained by combining all the $k-1$

eigenvectors. Finally, each row of Eig is regarded as the feature of a node, and the distance between two nodes can be obtained by calculating the Euclidean distance between the corresponding features.

2) *Random walk based distance*: Random walk is a stochastic process that starts from an initial node and goes to adjacent nodes with a probability [43]. Based on Markov chains, this process can be modeled by the average commute time (ACT) [44], which defines the time of a random walk starting from a node, arriving at another node for the first time, and going back to the node. As reported in [45], [46], the ACT can be used to measure the distance between any two nodes in a network. The ACT based distance between two nodes $ACT(v_i, v_j)$ can be calculated by [44]

$$ACT(v_i, v_j) = 2m(\mathbf{u}_i - \mathbf{u}_j)L^+(\mathbf{u}_i - \mathbf{u}_j)^T, \quad (12)$$

where $\mathbf{u}_i \in \mathbb{R}^{1 \times n}$ is a vector of zeros with only the i -th element being 1, and L^+ is the Moore-Penrose pseudoinverse of the Laplace matrix L , which can be obtained by [47]

$$L^+ = (L - \frac{II^T}{n})^{-1} + \frac{II^T}{n}, \quad (13)$$

where $I \in \mathbb{R}^{n \times 1}$ is a vector of ones.

3) *Diffusion kernel similarity*: Since the distance between two nodes is usually negatively related to the similarity between them, the similarity can be used to measure the distance between nodes in a network. In [48], the diffusion kernels on networks were tailored for kernel based learning algorithms. Diffusion kernels similarity is a kind of exponential kernels based on heat equation, which can be calculated by [48]

$$K = e^{-\beta \times L}, \quad (14)$$

where β is suggested to be 1 and L is the Laplace matrix. Note that since the diffusion kernels similarity is negatively related to the distance, we use the dissimilarity $DK = (DK_{ij})^{n \times n}$ to measure the distance between nodes, i.e.,

$$DK = \max(K) - K, \quad (15)$$

where $\max(K)$ denotes the maximum value in K , and the element DK_{ij} denotes the distance between nodes v_i and v_j .

B. Optimization Process of EMOFM

As shown in Fig. 2, the proposed EMOFM contains two stages: The first stage aims to uncover the nonoverlapping communities by finding the optimal central nodes of communities, while the second stage detects the overlapping nodes by finding the optimal fuzzy thresholds of non-central nodes. Both two stages are implemented based on an effective MOEA called AR-MOEA [24], but the objectives to be optimized in the two stages are different. Specifically, the first stage optimizes two objectives related to nonoverlapping communities, and the second stage optimizes two objectives related to overlapping communities.

The procedure of the proposed EMOFM is summarized in Algorithm 1. In the first stage, EMOFM generates an initial population P by an initialization strategy (Line 1), where each individual $\mathbf{p}_i$ in the population consists of a binary vector $\mathbf{p}_{i_b}$

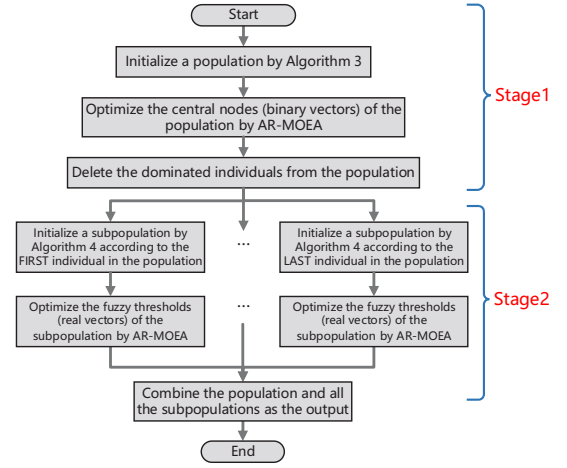


Fig. 2. Optimization process of EMOFM.

Algorithm 1: General Procedure of EMOFM

Input: Gen : Maximum number of generations for each stage;
 N_{pop} : Population size; N_{sub} : Subpopulation size;
Output: FR : Final result;
Stage 1: Uncovering nonoverlapping communities
 1: $P \leftarrow Initialization(N_{pop})$; // Algorithm 3
 2: $P \leftarrow AR_MOEA(P, Gen, 1)$; // Algorithm 2
 3: Delete all dominated individuals from P ;
Stage 2: Detecting overlapping nodes
 4: **for** $i = 1$ to $|P|$ **do**
 5: $SubP_i \leftarrow SubInitialization(\mathbf{p}_i, N_{sub})$; // Algorithm 4
 6: $SubP_i \leftarrow AR_MOEA(SubP_i, Gen, 2)$; // Algorithm 2
 7: Delete all dominated individuals from $SubP_i$;
 8: **end for**
 9: $FR \leftarrow SubP_1 \cup \dots \cup SubP_{|P|}$;
 10: **return** FR ;

and a real vector $\mathbf{p}_{i_r}$. Then, it optimizes the center nodes (represented by the binary vector) according to the following two widely used objectives for Gen generations (Line 2):

$$\min \begin{cases} KKM = 2(n - k) - \sum_{i=1}^k \frac{L(C_i, C_i)}{|C_i|} \\ RC = \sum_{i=1}^k \frac{L(C_i, V \setminus C_i)}{|C_i|} \end{cases}, \quad (16)$$

where n is the number of nodes of network $G(V, E)$, k is the number of communities, C_i denotes the i -th community, $L(C_i, C_j) = \sum_{v \in C_i, w \in C_j} A_{vw}$, and A is the adjacent matrix of the network. The first objective KKM (i.e., kernel k -means [26]) is the intra-link density in all communities, and the second objective RC (i.e., ratio cut [27]) is the inter-link density between different communities, hence it is expected to obtain a community structure with high intra-link density and low inter-link density by optimizing these two objectives. Note that the first stage aims to uncover the nonoverlapping communities, which are obtained by (8) according to each individual.

In the second stage, EMOFM first generates one subpopulation $SubP_i$ based on the i -th non-dominated individual in P (Line 5). Afterwards, for each subpopulation $SubP_i$, EMOFM optimizes the fuzzy thresholds (represented by the real vector)

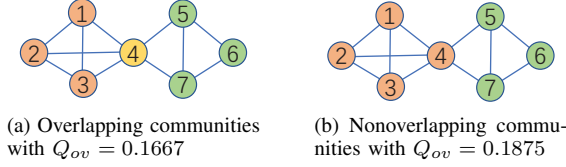


Fig. 3. Two community structures of a network, where the Q_{ov} value will increase if node 4 is regarded as a nonoverlapping node.

according to the following two objectives for Gen generations (Line 6):

$$\max \begin{cases} f_1 = Q_{ov} \\ f_2 = Num_{overlapping} \end{cases}, \quad (17)$$

where $Num_{overlapping}$ is the number of overlapping nodes, and Q_{ov} is the extended modularity [28]:

$$Q_{ov} = \frac{1}{2m} \sum_{i=1}^k \sum_{v \in C_i, w \in C_i} \frac{1}{O_v O_w} [A_{vw} - \frac{d_v d_w}{2m}], \quad (18)$$

where m is the number of edges in the network, O_v is the number of communities which node v belongs to, and d_v is the degree of node v . The larger the value of Q_{ov} , the better the quality of detected overlapping communities, hence it can be used to guide EMOFM to detect the overlapping nodes. The reason for maximizing the number of overlapping nodes is that not all meaningful overlapping nodes can be found by maximizing only Q_{ov} . Let us consider the network shown in Fig. 3, where node 4 is likely to be an overlapping node. However, a larger Q_{ov} value can be obtained if node 4 is regarded as a nonoverlapping node. Note that the overlapping communities are obtained by (9) for the calculation of Q_{ov} .

The optimization process of AR-MOEa is detailed in Algorithm 2. Different from the original AR-MOEa, the AR-MOEa adopted here optimizes different objective functions in the two stages. More specifically, it evaluates individuals by (16) in the first stage and by (17) in the second stage (Lines 1–5), and it evolves the binary vectors in the first stage and real vectors in the second stage (Lines 12–20). Besides, the mating selection (Line 7) is to select a set of parents with good convergence and diversity from P for generating offsprings, and the environmental selection (Line 23) is to retain some individuals with good convergence and diversity from $P \cup O$ for the next generation. Both the mating selection and environmental selection are based on an enhanced inverted generational distance based selection with adaptive reference points, the details of which are elaborated in [24].

Finally, as shown in Line 9 of Algorithm 1, EMOFM returns the population obtained in the first stage and all the subpopulations obtained in the second stage as the final result. In the next two subsections, the population initialization strategies in the two stages of EMOFM are elaborated, respectively.

C. Population Initialization Strategy in the First Stage

The population initialization strategy in the first stage aims to generate a set of initial individuals, such that some central nodes in these individuals have a large probability of being the centers of real communities. For this aim, we consider the

Algorithm 2: $AR_MOEA(P, Gen, stage)$

Input: P : Population to be optimized; Gen : Maximum number of generations; $stage$: Current stage number;
Output: P : Optimized population;
1: **if** $stage == 1$ **then**
2: Evaluate the individuals in P by (16);
3: **else if** $stage == 2$ **then**
4: Evaluate the individuals in P by (17);
5: **end if**
6: **for** $g = 1$ to Gen **do**
7: $P' \leftarrow MatingSelection(P)$;
8: $O \leftarrow \emptyset$;
9: **while** $P' \neq \emptyset$ **do**
10: Randomly select two individuals $\mathbf{p}_i$ and $\mathbf{p}_j$ from P' ;
11: $P' \leftarrow P' \setminus \{\mathbf{p}_i, \mathbf{p}_j\}$;
12: **if** $stage == 1$ **then**
13: $[\mathbf{b}_1, \mathbf{b}_2] \leftarrow$ Perform binary genetic operators on $\mathbf{p}_{i_b}$ and $\mathbf{p}_{j_b}$;
14: $\mathbf{p}_{i_b} \leftarrow \mathbf{b}_1, \mathbf{p}_{j_b} \leftarrow \mathbf{b}_2$;
15: Evaluate the modified $\mathbf{p}_i$ and $\mathbf{p}_j$ by (16);
16: **else if** $stage == 2$ **then**
17: $[\mathbf{r}_1, \mathbf{r}_2] \leftarrow$ Perform real genetic operators on $\mathbf{p}_{i_r}$ and $\mathbf{p}_{j_r}$;
18: $\mathbf{p}_{i_r} \leftarrow \mathbf{r}_1, \mathbf{p}_{j_r} \leftarrow \mathbf{r}_2$;
19: Evaluate the modified $\mathbf{p}_i$ and $\mathbf{p}_j$ by (17);
20: **end if**
21: $O \leftarrow O \cup \{\mathbf{p}_i, \mathbf{p}_j\}$;
22: **end while**
23: $P \leftarrow EnvironmentalSelection(P \cup O)$;
24: **end for**
25: **return** P ;

Algorithm 3: $Initialization(N_{pop})$

Input: N_{pop} : Population size;
Output: P : Initial population;
Step 1: Find candidate central nodes
1: $V \leftarrow \{1, 2, \dots, n\}$; // Set of remaining nodes
2: $K \leftarrow \emptyset$; // Set of candidate central nodes
3: **while** $V \neq \emptyset$ **do**
4: $w \leftarrow \operatorname{argmax}_{v \in V} d_v$; // d_v is the degree of node v
5: $K \leftarrow K \cup \{w\}$;
6: Delete w and all its adjacent nodes from V ;
7: **end while**
Step 2: Initialize the population
8: $P \leftarrow N_{pop}$ individuals with all variables being 0;
9: **for** $i = 1$ to $N_{pop}/2$ **do**
10: $rand \leftarrow$ A random integer within $[1, |K|]$;
11: Randomly select $rand$ nodes from K and set the corresponding binary variables of $\mathbf{p}_{i_b}$ to 1;
12: **end for**
13: **for** $i = N_{pop}/2 + 1$ to N_{pop} **do**
14: $rand \leftarrow$ A random integer within $[1, n]$;
15: Randomly select $rand$ nodes from $\{1, 2, \dots, n\}$ and set the corresponding binary variables of $\mathbf{p}_{i_b}$ to 1;
16: **end for**
17: **return** P ;

nodes with larger degrees as candidate central nodes, since the nodes in the center of a community are likely to have more connections to the others. Therefore, the nodes in the initial individuals with larger degrees are more likely to be central nodes (i.e., the corresponding binary variables are set to 1).

The detailed procedure of the population initialization strategy is shown in Algorithm 3. To find the candidate central nodes, the node with the maximum degree is selected as a candidate central node, and all its adjacent nodes are ignored. After that, the node with the maximum degree among the

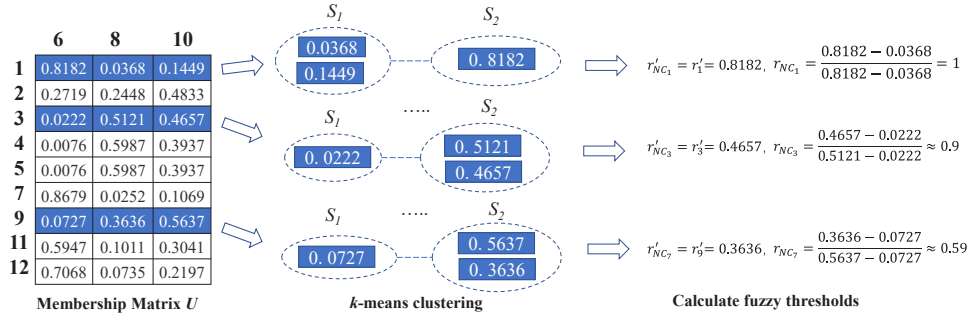


Fig. 4. An illustrative example of the initialization of fuzzy thresholds on the ENZYMES_g163 network. The values of the membership matrix are randomly generated for illustrative purposes.

remaining nodes is selected as a candidate central node, and this procedure repeats until all the nodes are selected or ignored (Lines 3–7).

Afterwards, the initial population P can be generated based on the candidate central nodes, where some candidate central nodes are randomly selected for each individual in P , and the values of corresponding binary variables of the individual are set to 1 (Lines 9–12). Note that in order to improve the diversity of the initial population as well as alleviate premature convergence, only half the individuals in P are generated by the above strategy, while the other half are generated by randomly selecting nodes in the network as central nodes (Lines 13–16).

D. Subpopulation Initialization Strategy in the Second Stage

After optimizing the center nodes in the first stage, the fuzzy thresholds will be optimized in the second stage. However, due to the difference between the center nodes in the optimized individuals, they cannot be optimized together in the second stage. In other words, the fuzzy thresholds of non-central nodes for each individual in P should be optimized separately. To this end, a subpopulation $SubP_i$ needs to be generated for each individual $\mathbf{p}_i$, so that the fuzzy thresholds of non-central nodes for $\mathbf{p}_i$ can be optimized by evolving $SubP_i$.

The detailed procedure of the subpopulation initialization strategy in the second stage is given in Algorithm 4, which consists of two steps, i.e., initialization of fuzzy thresholds and initialization of subpopulations. To initialize the fuzzy thresholds for individual $\mathbf{p}_i$, the membership matrix U between NC and CN is first calculated according to the binary vector $\mathbf{p}_{ib}$ (Line 2). Then, for each non-central node NC_j , the membership degrees between NC_j and all central nodes $U_{j1}, U_{j2}, \dots, U_{j|CN|}$ are clustered into two groups S_1 and S_2 , where the values in S_1 are smaller than those in S_2 (Line 5). Finally, the minimum value in S_2 is regarded as r'_{NC_j} , and r_{NC_j} can be calculated by the inverse operation of (10) (Line 6), i.e.,

$$r_{NC_j} = \frac{r'_{NC_j} - \min_l U_{jl}}{\max_l U_{jl} - \min_l U_{jl}}. \quad (19)$$

An example of initializing fuzzy thresholds on the ENZYMES_g163 network is given in Fig. 4, where the values of U are randomly generated for illustrative purposes.

Algorithm 4: *SubInitialization*($\mathbf{p}_i, N_{sub}$)

Input: $\mathbf{p}_i$: The i -th individual; N_{sub} : Subpopulation size;

Output: $SubP_i$: The i -th subpopulation;

Step1: Initialize fuzzy thresholds

- 1: $[NC, CN] \leftarrow$ Determine the set of non-central nodes and the set of central nodes by (6) according to $\mathbf{p}_{ib}$;
- 2: $U \leftarrow$ Calculate the membership matrix between NC and CN by (7);
- 3: $[r_1, r_2, \dots, r_n] \leftarrow 0$; // Initial fuzzy thresholds
- 4: **for** $j = 1$ to $|NC|$ **do**
- 5: $[S_1, S_2] \leftarrow$ Use k -means to cluster $U_{j1}, U_{j2}, \dots, U_{j|CN|}$ into two sets;
- 6: $r'_{NC_j} \leftarrow$ Calculate the fuzzy threshold of NC_j by (19);
- 7: **end for**
- 8: **Step2: Initialize subpopulations**
- 9: $SubP_i \leftarrow \emptyset$;
- 10: **for** $t = 1$ to N_{sub} **do**
- 11: $\mathbf{p}_{ir} \leftarrow [r_1, r_2, \dots, r_n]$;
- 12: **for** $j = 1$ to n **do**
- 13: $rand \leftarrow$ A random value within $[0, 1]$;
- 14: **if** $rand < 0.5$ **then**
- 15: Set the j -th real variable of $\mathbf{p}_{ir}$ to a random value within $[0, 1]$;
- 16: **end if**
- 17: $SubP_i \leftarrow SubP_i \cup \{\mathbf{p}_{ir}\}$;
- 18: **end for**
- 19: **return** $SubP_i$;

After initializing the fuzzy thresholds of non-central nodes for individual $\mathbf{p}_i$, the subpopulation $SubP_i$ can be generated, where the real vectors of all the individuals in $SubP_i$ are the same to the fuzzy thresholds $r_1, r_2, \dots, r_n$ (Line 10). To increase the diversity of $SubP_i$, each real variable of the individuals in $SubP_i$ is randomly perturbed with a probability of 0.5 (Lines 12–15). As a consequence, the individuals in the subpopulation $SubP_i$ have the same central nodes as individual $\mathbf{p}_i$, so these individuals can be evolved together. On the other hand, these individuals have diverse real vectors that are generated by perturbing the initial fuzzy thresholds, and better fuzzy thresholds for the non-central nodes are expected to be obtained by evolving these individuals.

E. Complexity Analysis

The time complexity of the proposed EMOFM is mainly determined by two components, i.e., the calculation of objective functions and the optimization process of AR-MOEA.

According to the definitions of KKM , RC and Q_{ov} , the time complexities for calculating them are $O(n+m)$, $O(n+m)$ and $O(n^2)$, respectively, where n is the number of nodes and m is the number of edges of the network. The time complexity of one generation of AR-MOEA is $O(N^3)$ [24], where N is the population size. Therefore, the worst time complexity of EMOFM in the first stage is $O((n+m)N_{pop}Gen) + O(N_{pop}^3Gen)$ and the worst time complexity of EMOFM in the second stage is $O(n^2N_{sub}GenN_{pop}) + O(N_{sub}^3GenN_{pop})$, where N_{pop} is the population size in the first stage, N_{sub} is the subpopulation size in the second stage, and Gen is the maximum number of generations for each stage. To summarize, the total time complexity of EMOFM is $O((n+m+N_{pop}^2+n^2N_{sub}+N_{sub}^3)N_{pop}Gen)$. Since $n \gg N_{pop}$, $n \gg N_{sub}$ and $n^2 \gg m$ in general, the time complexity of EMOFM can be considered as $O(n^2N_{sub}N_{pop}Gen)$.

On the other hand, the space complexity of the proposed EMOFM is mainly determined by three variables, i.e., the distance matrix between each two nodes, the population in the first stage, and the subpopulations in the second stage, the space complexities of which are $O(n^2)$, $O(nN_{pop})$ and $O(nN_{sub}N_{pop})$, respectively. Therefore, the space complexity of EMOFM is $O(n(n+N_{sub}N_{pop}))$.

IV. EMPIRICAL RESULTS

In this section, the performance of the proposed EMOFM is verified by comparing it with six state-of-the-art approaches for overlapping community detection, namely, the local maximum degree nodes based approach (LMD) [8], Zhang's approach [49], the nonnegative matrix factorization approach (NMF) [50], the improved multiobjective quantum-behaved particle swarm optimization (IMOQPSO) [51], the maximal clique based multiobjective evolutionary algorithm (MCMOEA) [52], and the mixed representation based multiobjective evolutionary algorithm (MR-MOEA) [12]. Among the six baseline approaches, LMD is based on local expansion, Zhang's approach is based on core nodes, NMF is based on fuzzy clustering, and IMOQPSO, MCMOEA, and MR-MOEA are based on evolutionary multiobjective optimization. In addition, we embed the three distance measures introduced in Section III-A in EMOFM, which are hereafter denoted as EMOFM-SC (with spectral clustering based Euclidean distance), EMOFM-RW (with random walk based distance), and EMOFM-DK (with diffusion kernel similarity), respectively.

In the rest of this section, the performance of EMOFM on real-world networks and LFR benchmark networks are verified in Section IV-B and Section IV-C, respectively, the effectiveness of the population initialization strategy in EMOFM is verified in Section IV-D, and the effectiveness of the two-stage optimization process in EMOFM is verified in Section IV-E. Lastly, the limitations of EMOFM are discussed in Section IV-F.

A. Experimental Settings

1) *Parameter settings of approaches*: The threshold controlling the merge of communities in LMD is set to 0.4, and the parameters in the other baseline approaches are set

TABLE III
CHARACTERISTICS OF 16 REAL-WORLD NETWORKS.

Network	Nodes	Edges	Avg. degree	Communities	Type
ENZYMES _{g163} [40] ¹	12	22	3.67	unknown	Cheminformatics
Karate [55] ²	34	78	4.69	2	Social
Dolphin [56] ²	62	159	5.13	2	Social
Polbook [57] ²	105	441	8.4	3	Social
Football [57] ²	115	613	10.66	12	Social
Email [21] ³	1133	5451	4.81	unknown	Social
Blogs [12] ⁴	3984	6803	3.41	unknown	Social
ego-Facebook [32] ⁶	4039	88234	21.85	unknown	Social
SFI [27] ⁴	118	200	1.69	unknown	Collaboration
Jazz [58] ³	198	2742	27.70	unknown	Collaboration
Netscience [27] ²	1589	2742	3.45	unknown	Collaboration
Erdős [59] ⁷	6927	11850	1.71	unknown	Collaboration
Celegans metabolic [59] ³	453	2040	4.50	unknown	Biology
Yeast-D2 [60] ⁵	1443	6993	9.69	162	Biology
Y2H [61] ⁵	1966	2705	2.75	203	Biology
PPI [9]	2445	6265	5.12	unknown	Biology

1. <http://networkrepository.com/chem.php>.
2. <http://www-personal.umich.edu/~mejn/netdata/>.
3. <http://deim.urv.cat/~alexandre.arenas/data/welcome.htm>.
4. <http://dsec.pku.edu.cn/~jliu/>.
5. <http://faculty.uaeu.ac.ae/nzaki/ProRank.htm>.
6. <http://snap.stanford.edu/data/>.
7. <http://vlado.fmf.uni-lj.si/pub/networks/pajek/data/gphs.htm>.

TABLE IV
PARAMETER SETTINGS OF LFR NETWORKS.

n	μ	O_n	O_m	c_{min}	c_{max}	Others
100	{0.1,0.2,0.3,0.4,0.5,0.6}	0.1n	{2,4,6,8}	5	12	$k=10$
500	{0.1,0.2,0.3,0.4,0.5,0.6}	0.1n	{2,4,6,8}	10	50	$k_{max}=50$
1000	{0.1,0.2,0.3,0.4,0.5,0.6}	0.1n	{2,4,6,8}	10	50	$\tau_1=2, \tau_2=1$

as suggested in their original literature. For EMOFM, the population size N_{pop} is set to 100, the subpopulation size N_{sub} is set to 10, the maximum number of generations Gen is set to 100. The uniform crossover and bitwise mutation are adopted in the first stage, and the simulated binary crossover [53] and polynomial mutation [54] are adopted in the second stage. The probability of crossover is set to 1, the probability of mutation is set to $1/n$ (n denotes the number of nodes in the network), and the distribution index of crossover and mutation is set to 20. As for the other MOEA based approaches, the population size and the number of function evaluations are set to the same to those in EMOFM. Besides, Table II lists the representations, genetic operators, and relevant parameter settings in all the MOEA based approaches.

2) *Real-world networks*: As shown in Table III, 16 well-known and widely used real-world networks are adopted in the experiments. These networks are extracted from various domains and with different scales, degree distributions, and characteristics, which can not only assess the performance of community detection approaches, but also challenge them in terms of robustness and scalability.

3) *LFR benchmark networks*: A number of synthetic networks produced by the Lancichinetti-Fortunato-Radicchi (LFR) model [62] are also adopted to evaluate the performance of the compared approaches. The LFR networks follow the settings in [52], which are listed in Table IV. Specifically, the number of nodes n is set to 100, 500 and 1000, the mixing parameter μ varies from 0.1 to 0.6, the overlapping membership O_m varies from 2 to 8, the number of overlapping

TABLE II
REPRESENTATIONS AND OPERATORS USED IN THE MOEA BASED APPROACHES.

Algorithm	Representation	Operators	Parameter settings
IMOQPSO	Line graph based representation	Improved real based quantum-behaved particle swarm optimization	Harmony memory considering rate is set to a random value in $[0.5, 0.9]$, pitch adjusting rate is set to a random value in $[0.3, 0.5]$, bandwidth is set to a random value in $[0.05, 0.1]$, and the number of measurements is set to 5
MC-MOEA	Maximal clique based representation	One-way crossover and improved label based mutation	Crossover probability is set to 0.7
MR-MOEA	Overlapping node based representation	Improved integer based particle swarm optimization	Inertia weight is set to a random value in $[0, 1]$ and both cognitive component and social component are set to 1.494
EMOFM	Central node and fuzzy threshold based representation	Uniform crossover, bitwise mutation, simulated binary crossover, and polynomial mutation	Crossover probability is set to 1, mutation probability is set to $1/n$, and distribution index of both crossover and mutation is set to 20 (n denotes the number of nodes)

nodes O_n is set to $0.1n$, the minimum number of nodes in each community is set to 5 and 10, the maximum number of nodes in each community is set to 12 and 50, and the other parameters k , k_{max} , τ_1 and τ_2 are fixed to 10, 50, 2 and 1, respectively. As a result, there are 72 LFR networks with different scales and characteristics.

4) *Performance metrics*: Two popular metrics are used to quantitatively compare the quality of overlapping communities obtained by different approaches on the benchmark networks, i.e., the extended modularity (Q_{ov}) [28] and the generalized normalized mutual information ($gNMI$) [63]. The definition of Q_{ov} can be found in (18), where a larger value of Q_{ov} indicates a better overlapping community structure. On the other hand, $gNMI$ is an information-theoretic measure of the agreement between two community structures, which is used to evaluate the agreement between the obtained communities and the ground truth. The $gNMI$ between a ground truth CA and the communities obtained by an approach CB can be mathematically defined as

$$gNMI(CA, CB) = \frac{-2 \sum_{i=1}^{k_{CA}} \sum_{j=1}^{k_{CB}} N_{ij} \log(\frac{N_{ij}}{N_{i.} N_{.j}})}{\sum_{i=1}^{k_{CA}} N_{i.} \log(\frac{N_{i.}}{n}) + \sum_{j=1}^{k_{CB}} N_{.j} \log(\frac{N_{.j}}{n})}, \quad (20)$$

where k_{CA} denotes the number of communities in CA , k_{CB} denotes the number of communities in CB , $N \in \mathbb{R}^{k_{CA} \times k_{CB}}$ is a matrix with N_{ij} being the number of nodes in the i -th community of CA that appear in the j -th community of CB , $N_{i.}$ is the sum over the i -th row of N , and $N_{.j}$ is the sum over the j -th column of N . A larger value of $gNMI$ indicates a higher similarity between CA and CB . It is worth noting that the evolutionary multiobjective optimization based approaches can obtain multiple overlapping community structures in a single run, we select the one having the best metric value as the final output as suggested in [27], [51], [52].

Each approach is executed for 30 independent runs on each network, and the best value, average value, and standard deviation of the metric values are recorded. All the experiments are carried out on a computer with Intel I5-6400 2.70 GHz CPU, 8 GB RAM, and Windows 10 operating system.

B. Results on Real-World Networks

The Q_{ov} values obtained by six baseline approaches and three EMOFM based approaches on 16 real-world networks are listed in Table V. Note that the results of Zhang's approach, IMOQPSO, and MCMOEA on some networks cannot be obtained due to the limitation of internal storage. As can

TABLE VI
FRIEDMAN TEST WITH NEMENYI PROCEDURE ON AVERAGE Q_{ov} OBTAINED BY MR-MOEA AND THREE EMOFM BASED APPROACHES, WHERE '1' INDICATES SIGNIFICANT DIFFERENCE AND '0' OTHERWISE.

Algorithm	MR-MOEA	EMOFM-RW	EMOFM-SC	EMOFM-DK
MR-MOEA	NaN	1	1	1
EMOFM-RW	1	NaN	0	0
EMOFM-SC	1	0	NaN	0
EMOFM-DK	1	0	0	NaN
Friedman rank	1.0938	2.4688	3.0313	3.4063
p -value	5.2679e-07	significance level α		0.05

be seen, the fuzzy clustering based approach NMF and the MOEA based approach MR-MOEA exhibit relatively good performance among the six baseline approaches, whereas they underperform the EMOFM based approaches that combine both fuzzy clustering and MOEA. Moreover, the Friedman test with Nemenyi procedure [64] is conducted on the average Q_{ov} of MR-MOEA and the three EMOFM based approaches, which is a nonparametric statistical procedure for checking whether a set of samples are statistically different. According to Table VI, the p -value smaller than the significance level α indicates that the performance of the four approaches is significantly different, where MR-MOEA is statistically different from the EMOFM based approaches and the EMOFM based approaches are statistically similar to each other. Besides, the Friedman ranks indicate that MR-MOEA has worse performance than the EMOFM based approaches, since a smaller Friedman rank indicates a smaller Q_{ov} . As a result, the superiority of the proposed EMOFM over existing approaches for overlapping community detection is confirmed.

Table VII shows the $gNMI$ values obtained by EMOFM and the compared approaches on the real-world networks whose ground truth is known¹. It can be observed from the table that the EMOFM based approaches also perform the best on 5 out of 6 networks, while their $gNMI$ values are slightly smaller than those of NMF, IMOQPSO, and MR-MOEA on the Polbook network. This is due to the inconsistency between the metrics Q_{ov} and $gNMI$ [65], and the EMOFM based approaches can obtain better results in terms of Q_{ov} since they directly adopt Q_{ov} as an objective to be optimized.

To take a close look at the results obtained by different approaches, Table VIII presents the numbers of communities

¹The ground truths of Karate, Dolphin, Polbook, and Football networks can be obtained from <http://web.xidian.edu.cn/rhshang/paper.html>, and the ground truths of Yeast-D2 and Y2H networks can be obtained from <http://faculty.uaeu.ac.ae/nzaki/ProRank.htm>

TABLE V
STATISTICAL RESULTS OF Q_{ov} OBTAINED BY SIX BASELINE APPROACHES AND THREE EMOFM BASED APPROACHES ON 16 REAL-WORLD NETWORKS. THE BEST RESULT IN EACH NETWORK IS HIGHLIGHTED.

Network	Metric	LMD	Zhang	NMF	IMOQPSO	MCMOEA	MR-MOEA	EMOFM-SC	EMOFM-RW	EMOFM-DK
ENZYMES_g163	Q_{ov_Max}	0.2397	0.2438	0.1963	0.2438	0.2438	0.2567	0.2567	0.2567	0.2567
	Q_{ov_Avg}	0.2397	0.2438	0.1963	0.2431	0.2432	0.2567	0.2567	0.2567	0.2567
	Std	0	0	0	0.0023	0.0035	0	0	0	0
Karate	Q_{ov_Max}	0.2162	0.1900	0.2081	0.2130	0.2107	0.2285	0.2348	0.2348	0.2348
	Q_{ov_Avg}	0.2145	0.1900	0.2081	0.2080	0.2087	0.2261	0.2346	0.2325	0.2341
	Std	0.0022	0	0	0.0040	0.0017	0.0039	0.0001	0.0027	0.0008
Dolphin	Q_{ov_Max}	0.2614	0.2349	0.2636	0.2088	0.2212	0.2626	0.2718	0.2741	0.2730
	Q_{ov_Avg}	0.2600	0.2236	0.2636	0.2040	0.2024	0.2597	0.2700	0.2717	0.2723
	Std	0.0009	0.0062	0.0000	0.0028	0.0103	0.0022	0.0008	0.0016	0.0008
Polbook	Q_{ov_Max}	0.2689	0.2544	0.2591	0.2289	0.2379	0.2649	0.2702	0.2704	0.2704
	Q_{ov_Avg}	0.2554	0.2515	0.2591	0.2269	0.2179	0.2631	0.2701	0.2702	0.2703
	Std	0.0054	0.002	0	0.0022	0.0136	0.0015	0.0001	0.0002	0.0001
Football	Q_{ov_Max}	0.2982	0.2889	0.3049	0.2431	0.2785	0.3049	0.3067	0.3067	0.3067
	Q_{ov_Avg}	0.2917	0.2874	0.3049	0.2352	0.2726	0.3035	0.3066	0.3063	0.3066
	Std	0.0031	0.0019	0.0000	0.0143	0.0064	0.0018	0.0000	0.0003	0.0001
Email	Q_{ov_Max}	0.2287	0.2102	0.2368	0.1170	0.0826	0.2463	0.2799	0.2783	0.2823
	Q_{ov_Avg}	0.2234	0.2033	0.2352	0.1132	0.0774	0.2333	0.2779	0.2723	0.2784
	Std	0.0031	0.0025	0.0024	0.0023	0.0029	0.0069	0.0022	0.0031	0.0028
Blogs	Q_{ov_Max}	0.3591	0.3429	0.3991	0.3561	0.3347	0.3955	0.4173	0.4013	0.4165
	Q_{ov_Avg}	0.3555	0.3297	0.3967	0.3534	0.3297	0.3875	0.4137	0.3977	0.4131
	Std	0.0021	0.0025	0.0023	0.0023	0.0025	0.0027	0.0020	0.0028	0.0018
ego-Facebook	Q_{ov_Max}	0.4041	-	0.3520	-	-	0.4125	0.4111	0.4166	0.4145
	Q_{ov_Avg}	0.4029	-	0.3492	-	-	0.4064	0.4090	0.4151	0.4126
	Std	0.0013	-	0.0028	-	-	0.0037	0.0021	0.0015	0.0010
SFI	Q_{ov_Max}	0.3496	0.3173	0.3765	0.3283	0.2285	0.3741	0.3851	0.3841	0.3851
	Q_{ov_Avg}	0.3459	0.3032	0.3765	0.3242	0.1976	0.3672	0.3821	0.3826	0.3843
	Std	0.0026	0.0050	0.0000	0.0022	0.0131	0.0040	0.0013	0.0010	0.0010
Jazz	Q_{ov_Max}	0.1485	0.1482	0.1653	0.1562	0.1458	0.2190	0.2259	0.2259	0.2259
	Q_{ov_Avg}	0.1465	0.1394	0.1653	0.0873	0.1324	0.2169	0.2258	0.2259	0.2258
	Std	0.0008	0.0132	0.0000	0.0514	0.0091	0.0023	0.0002	0.0001	0.0001
Netscience	Q_{ov_Max}	0.4263	0.3196	0.4655	0.3623	0.4522	0.4612	0.4718	0.4713	0.4733
	Q_{ov_Avg}	0.4212	0.3195	0.4496	0.3592	0.4482	0.4563	0.4698	0.4696	0.4709
	Std	0.0022	0.0001	0.0101	0.0034	0.0022	0.0109	0.0009	0.0012	0.0013
Erdős	Q_{ov_Max}	0.2824	-	0.3126	-	-	0.3140	0.3349	0.3166	0.3315
	Q_{ov_Avg}	0.2822	-	0.3093	-	-	0.2970	0.3329	0.3086	0.3283
	Std	0.0021	-	0.0035	-	-	0.0108	0.0019	0.0055	0.0019
Celegansmetabolic	Q_{ov_Max}	0.1401	0.0608	0.0968	0.0490	0.0336	0.0778	0.2211	0.2187	0.2187
	Q_{ov_Avg}	0.1352	0.0594	0.0911	0.0456	0.0284	0.0726	0.2178	0.2154	0.2171
	Std	0.0028	0.0015	0.0042	0.0018	0.0039	0.0026	0.0020	0.0023	0.0013
Yeast-D2	Q_{ov_Max}	0.3972	0.3912	0.2221	0.3020	0.3502	0.4132	0.4147	0.4144	0.4193
	Q_{ov_Avg}	0.3944	0.3893	0.2100	0.2541	0.3351	0.4081	0.4139	0.4115	0.4179
	Std	0.0015	0.0071	0.0067	0.0392	0.0100	0.0037	0.0013	0.0019	0.0010
Y2H	Q_{ov_Max}	0.2970	0.2519	0.3098	0.2810	0.2362	0.3223	0.3523	0.3455	0.3661
	Q_{ov_Avg}	0.2943	0.2508	0.3085	0.2810	0.2274	0.3152	0.3483	0.3423	0.3628
	Std	0.0008	0.0004	0.008	0	0.005	0.0055	0.0020	0.0021	0.0028
PPI	Q_{ov_Max}	0.2911	0.2569	0.3303	0.2562	0.2072	0.3235	0.3500	0.3377	0.3482
	Q_{ov_Avg}	0.2875	0.2561	0.3278	0.2510	0.1990	0.3193	0.3468	0.3304	0.3460
	Std	0.0017	0.0006	0.0014	0.0052	0.0053	0.0036	0.0023	0.0037	0.0017

TABLE VII
STATISTICAL RESULTS OF $gNMI$ OBTAINED BY SIX BASELINE APPROACHES AND THREE EMOFM BASED APPROACHES ON 6 REAL-WORLD NETWORKS. THE BEST RESULT IN EACH NETWORK IS HIGHLIGHTED.

Network	Metric	LMD	Zhang	NMF	IMOQPSO	MCMOEA	MR-MOEA	EMOFM-SC	EMOFM-RW	EMOFM-DK
Karate	$gNMI_Max$	0.5135	0.3477	0.4404	0.7081	0.9186	1	1	1	1
	$gNMI_Avg$	0.398	0.3477	0.4404	0.6982	0.8044	0.9837	1	1	1
	Std	0.0741	0	0	0.0241	0.1194	0.0517	0.0000	0.0000	0.0000
Dolphin	$gNMI_Max$	0.6110	0.3546	0.4664	0.7944	0.3134	1	0.8889	1	1
	$gNMI_Avg$	0.4562	0.3456	0.4664	0.7931	0.2734	0.9667	0.8889	0.9667	1
	Std	0.1321	0.0138	0.0000	0.0042	0.0277	0.0537	0.0000	0.0537	0.0000
Polbook	$gNMI_Max$	0.2295	0.1790	0.3880	0.4331	0.3079	0.4587	0.3496	0.3514	0.3371
	$gNMI_Avg$	0.2042	0.1783	0.3880	0.4065	0.2742	0.4228	0.3495	0.3395	0.3358
	Std	0.0311	0.0005	0	0.0193	0.0233	0.0407	0.0001	0.0063	0.0009
Football	$gNMI_Max$	0.7702	0.7119	0.8032	0.8092	0.7282	0.8030	0.8524	0.8565	0.8376
	$gNMI_Avg$	0.7230	0.7082	0.8032	0.7984	0.7097	0.8030	0.8129	0.8285	0.8245
	Std	0.0306	0.0048	0.0000	0.0153	0.0157	0	0.0179	0.0164	0.0130
Yeast-D2	$gNMI_Max$	0.2117	0.1844	0.2078	0.2052	0.2658	0.2641	0.2669	0.2625	0.5401
	$gNMI_Avg$	0.2050	0.1824	0.1968	0.2013	0.2548	0.2542	0.2667	0.2587	0.3162
	Std	0.0034	0.0013	0.0066	0.0041	0.0065	0.0047	0.0004	0.0037	0.1180
Y2H	$gNMI_Max$	0.0183	0.0313	0.0261	0.0184	0.0273	0.1182	0.1061	0.1383	0.0885
	$gNMI_Avg$	0.0183	0.0313	0.0261	0.0183	0.0227	0.1154	0.1006	0.1178	0.0856
	Std	0	0	0	0.001	0.0033	0.0019	0.0032	0.0102	0.0014

and overlapping nodes found by EMOFM and the compared approaches, where each cell shows the minimum and maximum numbers of communities or overlapping nodes found by an approach in 30 runs. Owing to the population based search mechanism of MOEA based approaches, the numbers of

communities and overlapping nodes found by them have much larger ranges than those found by LMD, Zhang's approach, and NMF. As for the EMOFM based approaches, the numbers of communities found by them have larger ranges than those found by the other approaches besides MR-MOEA, and the

TABLE VIII

MINIMUM NUMBER OF COMMUNITIES C_{Min} , MAXIMUM NUMBER OF COMMUNITIES C_{Max} , MINIMUM NUMBER OF OVERLAPPING NODES O_{Min} , AND MAXIMUM NUMBER OF OVERLAPPING NODES O_{Max} FOUND BY SIX BASELINE APPROACHES AND THREE EMOFM BASED APPROACHES ON 16 REAL-WORLD NETWORKS.

Network	Metric	LMD	Zhang	NMF	IMOQPSO	MCMOEa	MR-MOEa	EMOFM-SC	EMOFM-RW	EMOFM-DK
ENZYMES_g163	(C_{Min}, C_{Max})	(2,2)	(2,2)	(3,3)	(1,6)	(2,4)	(1,12)	(1,12)	(1,12)	(1,12)
	(O_{Min}, O_{Max})	(0,0)	(1,1)	(4,4)	(0,10)	(1,5)	(0,1)	(0,6)	(0,6)	(0,6)
Karate	(C_{Min}, C_{Max})	(3,4)	(4,4)	(4,4)	(3,7)	(2,6)	(1,32)	(1,30)	(1,29)	(1,28)
	(O_{Min}, O_{Max})	(3,4)	(8,8)	(8,8)	(2,31)	(1,10)	(0,3)	(0,26)	(0,24)	(0,25)
Dolphin	(C_{Min}, C_{Max})	(3,6)	(6,8)	(2,2)	(2,7)	(6,17)	(2,49)	(1,35)	(1,36)	(1,38)
	(O_{Min}, O_{Max})	(5,12)	(13,20)	(2,2)	(7,34)	(15,34)	(0,13)	(0,42)	(0,47)	(0,45)
Polbook	(C_{Min}, C_{Max})	(2,6)	(6,6)	(12,12)	(1,7)	(4,14)	(3,95)	(1,51)	(1,48)	(1,55)
	(O_{Min}, O_{Max})	(1,16)	(17,19)	(61,61)	(0,92)	(14,58)	(0,11)	(0,86)	(0,81)	(0,83)
Football	(C_{Min}, C_{Max})	(8,12)	(11,11)	(12,12)	(3,9)	(12,18)	(2,97)	(1,67)	(1,63)	(1,64)
	(O_{Min}, O_{Max})	(4,17)	(12,14)	(1,1)	(46,85)	(1,21)	(4,16)	(0,95)	(0,100)	(0,97)
Email	(C_{Min}, C_{Max})	(40,52)	(60,66)	(12,12)	(2,7)	(228,282)	(8,895)	(1,348)	(2,324)	(2,377)
	(O_{Min}, O_{Max})	(321,405)	(518,566)	(343,385)	(4,700)	(608,661)	(95,145)	(0,906)	(0,1109)	(0,1046)
Blogs	(C_{Min}, C_{Max})	(346,357)	(361,366)	(10,10)	(3,7)	(366,439)	(213,3480)	(2,968)	(1,961)	(2,938)
	(O_{Min}, O_{Max})	(461,531)	(761,804)	(252,296)	(200,384)	(690,746)	(217,298)	(0,3152)	(0,3455)	(0,1839)
ego-Facebook	(C_{Min}, C_{Max})	(59,65)	-	(117,156)	-	-	(8,4001)	(2,860)	(2,651)	(2,961)
	(O_{Min}, O_{Max})	(189,307)	-	(249,351)	-	-	(13,31)	(0,3100)	(0,3986)	(0,3964)
SFI	(C_{Min}, C_{Max})	(12,14)	(12,12)	(16,16)	(1,6)	(25,43)	(9,109)	(1,59)	(1,57)	(1,61)
	(O_{Min}, O_{Max})	(2,8)	(10,19)	(14,14)	(0,14)	(22,37)	(1,9)	(0,56)	(0,71)	(0,50)
Jazz	(C_{Min}, C_{Max})	(2,3)	(5,5)	(4,4)	(2,7)	(5,17)	(2,158)	(1,75)	(1,67)	(1,78)
	(O_{Min}, O_{Max})	(0,25)	(4,7)	(5,5)	(25,179)	(20,103)	(0,129)	(0,189)	(0,194)	(0,190)
Netscience	(C_{Min}, C_{Max})	(437,446)	(572,572)	(12,12)	(2,4)	(303,335)	(305,1376)	(2,413)	(2,414)	(2,396)
	(O_{Min}, O_{Max})	(20,36)	(388,389)	(3,12)	(2,1461)	(45,79)	(27,56)	(0,1336)	(0,944)	(0,646)
Erdős	(C_{Min}, C_{Max})	(380,386)	-	(22,39)	-	-	(52,6324)	(2,1805)	(2,1705)	(2,1732)
	(O_{Min}, O_{Max})	(1233,1291)	-	(774,909)	-	-	(264,361)	(0,5740)	(0,6610)	(0,4661)
Celegans metabolic	(C_{Min}, C_{Max})	(9,16)	(7,9)	(11,11)	(2,9)	(246,274)	(196,359)	(1,140)	(1,102)	(1,153)
	(O_{Min}, O_{Max})	(0,91)	(197,210)	(40,41)	(215,382)	(116,137)	(31,62)	(0,319)	(0,439)	(0,430)
Yeast-D2	(C_{Min}, C_{Max})	(143,152)	(213,218)	(172,186)	(3,5)	(155,196)	(106,1313)	(2,400)	(2,357)	(1,410)
	(O_{Min}, O_{Max})	(48,83)	(693,737)	(462,509)	(7,145)	(207,317)	(48,87)	(0,1229)	(0,1208)	(0,1018)
Y2H	(C_{Min}, C_{Max})	(447,458)	(502,507)	(203,203)	(4,4)	(418,482)	(244,1466)	(2,612)	(2,520)	(2,529)
	(O_{Min}, O_{Max})	(223,246)	(516,533)	(261,284)	(78,1966)	(554,611)	(216,298)	(0,1325)	(0,1712)	(0,1132)
PPI	(C_{Min}, C_{Max})	(202,217)	(251,257)	(237,245)	(3,6)	(355,422)	(79,1923)	(2,659)	(2,630)	(2,681)
	(O_{Min}, O_{Max})	(475,551)	(884,917)	(310,324)	(30,563)	(909,992)	(222,310)	(0,1993)	(0,2375)	(0,1520)

numbers of overlapping nodes found by them have larger ranges than those found by all the other approaches. As a result, the proposed EMOFM can find a number of diverse community structures with different numbers of communities and overlapping nodes.

According to Tables V and VII, although none of the three EMOFM based approaches can obtain the best results on all the real-world networks, they outperform the six baseline approaches in most cases. Moreover, their metric values are very similar to each other, which indicates that the performance of the proposed EMOFM is not sensitive to the distance measure used. For simplicity, we use EMOFM-DK as a representative of the proposed EMOFM in the subsequent experiments.

C. Results on LFR Benchmark Networks

The average $gNMI$ values obtained by six baseline approaches and EMOFM-DK on 72 LFR networks are shown in Fig. 5, where two observations can be made. First, EMOFM-DK outperforms the other approaches when the mixing parameter μ is small. This is because a small μ indicates an unambiguous community structure, which enables EMOFM-DK to find the central nodes easily. Second, EMOFM-DK is superior over the other approaches especially when the overlapping membership O_m is large, since EMOFM-DK assigns a fuzzy threshold to each node, which is helpful to the detection of nodes belonging to multiple communities. As a consequence, the proposed EMOFM also exhibits the best overall performance on the LFR networks considered here.

TABLE IX

AVERAGE Q_{ov} OBTAINED BY THE FIRST STAGES OF EMOFM-DK AND EMOFM-DK WITH RANDOM INITIALIZATION AND THREE MOEA BASED APPROACHES ON 14 REAL-WORLD NETWORKS. THE BEST RESULT IN EACH NETWORK IS HIGHLIGHTED.

Network	First stage of EMOFM-DK	First stage of EMOFM-DK with random initialization	IMOQPSO	MCMOEa	MR-MOEa
ENZYMES_g163	0.2567	0.2567	0.2431	0.2432	0.2567
Karate	0.2315	0.2258	0.2080	0.2087	0.2261
Dolphin	0.2634	0.2359	0.2040	0.2024	0.2597
Polbook	0.2667	0.1836	0.2269	0.2179	0.2631
Football	0.2948	0.1861	0.2352	0.2726	0.3035
Email	0.2539	0.0856	0.1132	0.0774	0.2333
Blogs	0.3872	0.1823	0.3534	0.3297	0.3875
SFI	0.3741	0.3054	0.3242	0.1976	0.3672
Jazz	0.2148	0.0479	0.0873	0.1324	0.2169
Netscience	0.4487	0.2437	0.3592	0.4482	0.4563
Celegans metabolic	0.2005	0.1205	0.0456	0.0284	0.0726
Yeast-D2	0.4008	0.1096	0.2541	0.3351	0.4081
Y2H	0.3380	0.1906	0.2810	0.2274	0.3152
PPI	0.3247	0.1409	0.2510	0.1990	0.3193

D. Effectiveness of the Population Initialization Strategy in EMOFM

We design two population initialization strategies in the proposed EMOFM, where the initialization strategy in the first stage can improve the quality of initial individuals, and the initialization strategy in the second strategy is necessary for generating a subpopulation for each individual. Since the initialization strategy in the first stage can be removed from EMOFM, we verify the effectiveness of the initialization strategy in the first stage by comparing it with the general initialization strategy, i.e., random initialization.

The average Q_{ov} values obtained by the first stages (i.e., finding only nonoverlapping communities) of EMOFM-DK

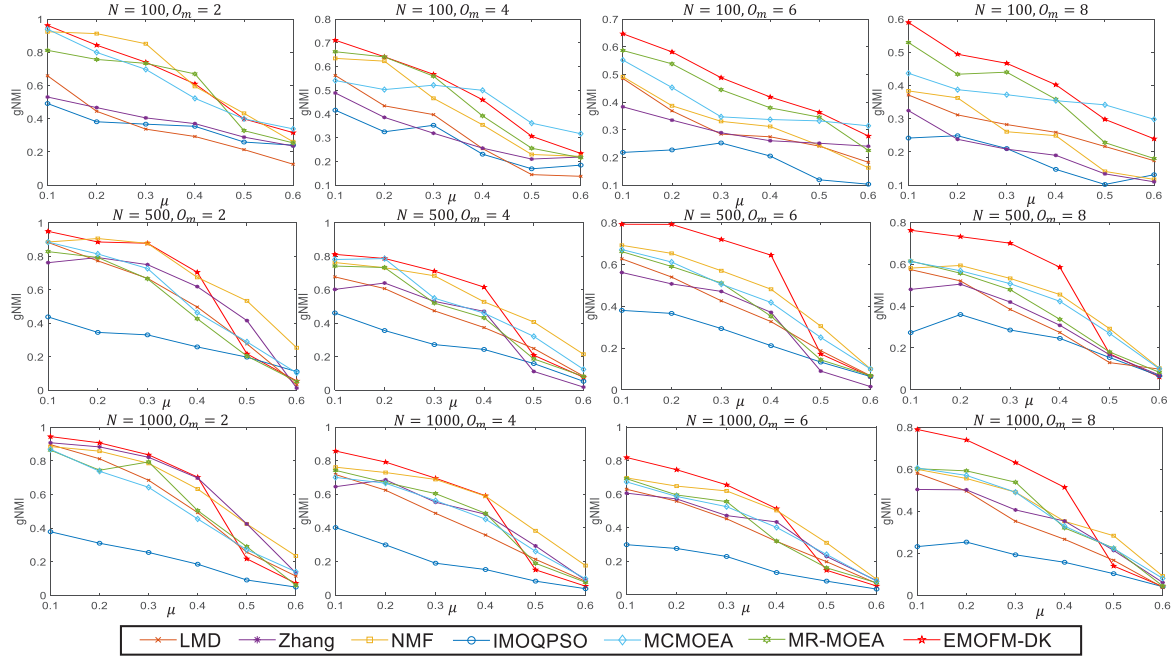


Fig. 5. Average $gNMI$ obtained by six baseline approaches and EMOFM-DK on 72 LFR networks.

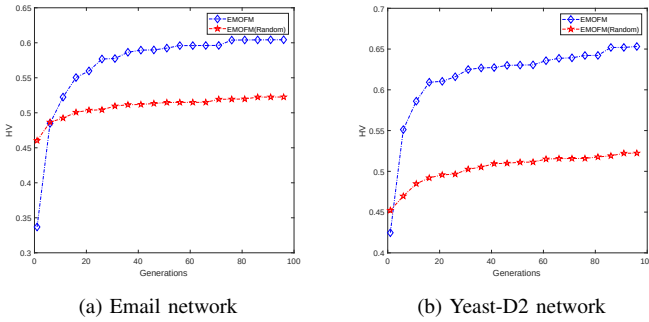


Fig. 6. Convergence profiles of HV obtained by EMOFM-DK and EMOFM-DK with random initialization strategy on two benchmark networks.

and EMOFM-DK with random initialization on 14 real-world networks can be found in Table IX. As can be observed, EMOFM-DK outperforms EMOFM-DK with random initialization on all the networks. For further observations, Fig. 6 depicts the convergence profiles of hypervolume (HV) [66] obtained by the two approaches on the Email network and the Yeast-D2 network, averaged over 30 runs. Given a population, the HV value is defined as the area covered by all the individuals in the population with respect to a reference point in objective space, where a larger HV value indicates a better convergence and diversity of the population. As shown in Fig. 6, the population generated by EMOFM-DK is worse than that generated by EMOFM-DK with random initialization in terms of HV, however, the former leads to a much faster convergence speed than the latter. Therefore, the proposed initialization strategy can significantly improve the performance of EMOFM.

TABLE X
FRIEDMAN TEST WITH NEMENYI PROCEDURE ON AVERAGE Q_{ov} OBTAINED BY THE FIRST STAGES OF EMOFM-DK, EMOFM-DK WITH RANDOM INITIALIZATION, AND THREE MOEA BASED APPROACHES, WHERE '1' INDICATES SIGNIFICANT DIFFERENCE AND '0' OTHERWISE.

Algorithm	First stage of EMOFM-DK	First stage of EMOFM-DK with random initialization	IMOQPSO	MCMOEa	MR-MOEa
First stage of EMOFM-DK	NaN	1	1	1	0
First stage of EMOFM-DK with random initialization	1	NaN	0	0	1
IMOQPSO	1	0	NaN	0	1
MCMOEa	1	0	0	NaN	1
MR-MOEa	0	1	1	1	NaN
Friedman rank	4.6071	1.8929	2.2857	2.0000	4.2143
p -value	1.0291e-07		significance level α		0.05

E. Effectiveness of the Two-Stage Optimization Process in EMOFM

To verify the effectiveness of the nonoverlapping community detection in the first stage of EMOFM, we compare the average Q_{ov} values obtained by the first stage of EMOFM-DK with those obtained by IMOQPSO, MCMOEa, and MR-MOEa. According to the results presented in Table IX, it can be found that the first stage of EMOFM-DK outperforms the other MOEA based approaches on 8 out of 14 real-world networks. Besides, Table X presents the statistical results obtained by Friedman test with Nemenyi procedure, it can be seen that the performance of the first stage of EMOFM-DK is statistically different from the performance of IMOQPSO and MCMOEa, and statistically similar to the performance of MR-MOEa. Furthermore, Fig. 7 plots the populations with the median HV value among 30 runs obtained by the four MOEA based approaches on the Karate network and the Dolphin

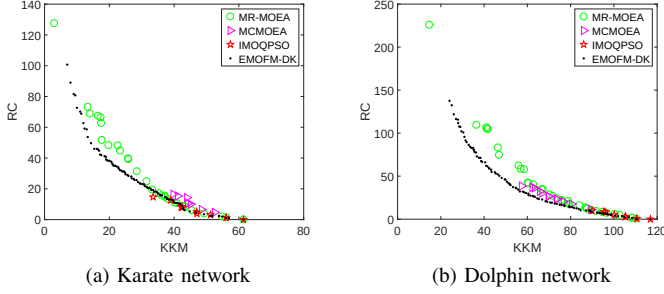


Fig. 7. Populations obtained by IMOQPSO, MCMOE, MR-MOE, and the first stage of EMOFM-DK on 2 benchmark networks.

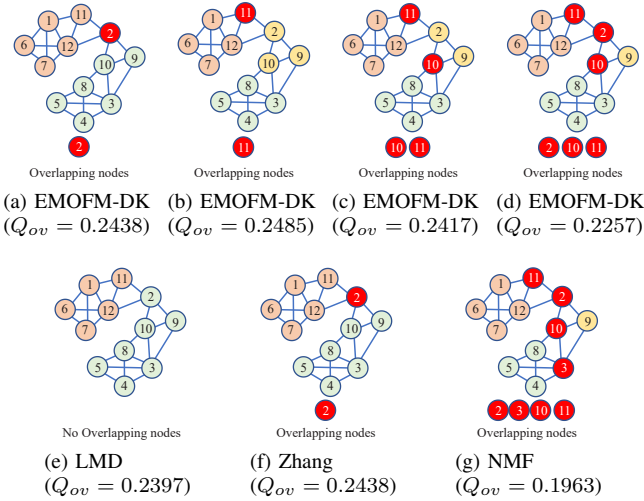


Fig. 8. Four overlapping community structures obtained by EMOFM-DK and the overlapping communities obtained by LMD, Zhang's approach, and NMF on the ENZYMES_g163 network.

network. As can be seen from Fig. 7, the community structures obtained in the first stage of EMOFM-DK have significantly better KKM and RC values than those obtained by the other MOEA based approaches. In fact, the inferiority of the other approaches is mainly attributed to the prior knowledge they rely on, which may lead to unexpected results if the prior knowledge is not ideal.

Lastly, we verify the effectiveness of the overlapping community detection in the second stage of EMOFM. Figs. 8 and 9 depict the overlapping communities obtained by EMOFM-DK, LMD, Zhang's approach, and NMF on the ENZYMES_g163 network and the Karate network, respectively. As can be observed, the overlapping communities found by EMOFM-DK shown in Fig. 8(a) is the same to that found by Zhang's approach, and the overlapping communities found by EMOFM-DK shown in Fig. 8(b) is with better Q_{ov} ; besides, EMOFM-DK can find more overlapping nodes as shown in Figs. 8(c)–(d). Similarly, the overlapping communities found by EMOFM-DK shown in Figs. 9(a)–(c) are with better Q_{ov} than those found by LMD, Zhang's approach, and NMF. To summarize, the proposed EMOFM can not only find the overlapping communities similar to those detected by the other approaches, but also detect more promising overlapping

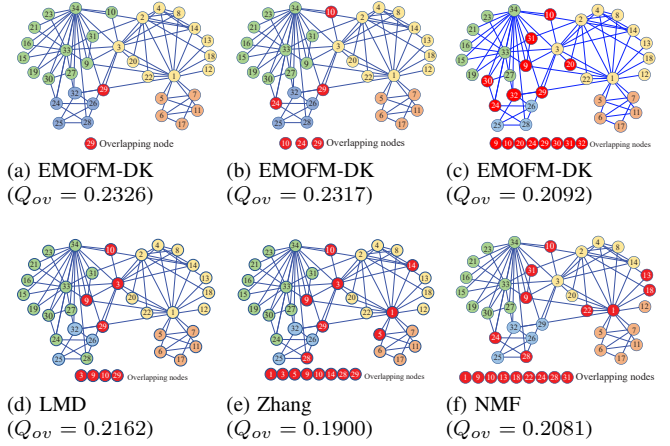


Fig. 9. Three overlapping community structures obtained by EMOFM-DK and the overlapping communities obtained by LMD, Zhang's approach, and NMF on the Karate network.

TABLE XI
AVERAGE RUNTIME OF EMOFM-DK AND THREE MOEA BASED APPROACHES ON 16 REAL-WORLD NETWORKS.

Network	Nodes	Runtime			
		IMOQPSO	MCMOE	MR-MOE	EMOFM-DK
ENZYMES_g163	12	4.42s	0.72s	8.63s	2.55s
Karate	34	5.32s	1.21s	19.34s	3.62s
Dolphin	62	6.96s	2.6s	56.70s	5.32s
Polbook	105	7.86s	4.07s	77.12s	8.13s
Football	115	8.71s	4.52s	93.72s	9.26s
SFI	118	8.46s	4.31s	66.48s	9.19s
Jazz	198	366.61s	13s	173.72s	16.69s
Celegansmetabolic	453	186.67s	39.90s	637.92s	48.16s
Email	1133	773.02s	177.62s	1529.96s	299.05s
Yeast-D2	1443	1895.67	282.61s	1805.83s	605.15s
Netscience	1589	538.63s	108.37s	1779.51s	488.78s
Y2H	1966	638.56s	506.94s	6647.14s	1238.60s
PPI	2445	1505.99s	944.05s	8069.49s	1862.29s
Blogs	3984	2758.92s	979.38s	15624.02s	3230.36s
ego-Facebook	4039	-	-	5953.07s	4377.45s
Erdős	6927	-	-	26287.01s	10937.85s

communities with different overlapping degrees.

F. Discussion

The above experimental results have demonstrated the effectiveness of the proposed EMOFM on overlapping community detection, while they also reveal the limitations of EMOFM. Firstly, according to the results on LFR benchmark networks shown in Fig. 5, the performance of EMOFM deteriorates considerably when the community structure is not distinct, since the central nodes of ambiguous communities are hard to be detected and thus the membership among nodes is inaccurate. Secondly, according to Fig. 8(g), node 3 is likely to be an overlapping node, but it is missed by EMOFM since it is detected as a central node and can never be an overlapping node due to the representation of EMOFM. Besides, according to the analysis in Section III-E, the time complexity of EMOFM increases considerably with the increase of number of nodes n , which is a common limitation of MOEA based approaches. Table XI lists the average runtime of EMOFM-DK and the other MOEA based approaches on 16 real-world networks, it can be found that the runtimes on the Erdős network with 6927 nodes is much more than that on the ENZYMES_g163 network with 12 nodes.

V. CONCLUSIONS

In this paper, we have proposed an evolutionary multi-objective optimization based fuzzy method for overlapping community detection. The proposed EMOFM enhances the performance of fuzzy clustering by optimizing the community centers via a well tailored MOEA, and it also optimizes the fuzzy thresholds for obtaining diverse overlapping community structures. Furthermore, to improve the performance of the MOEA in optimizing community centers and fuzzy thresholds, a two-stage optimization process and two population initialization strategies are proposed for EMOFM.

In the experiments, we have implemented three approaches by embedding three distance measures in EMOFM. The statistical results have indicated that EMOFM outperforms six state-of-the-art approaches for overlapping community detection, including fuzzy clustering based approaches and MOEA based approaches. In addition, the effectiveness of the population initialization strategy and the two-stage optimization process in EMOFM have also been verified.

This work has demonstrated the potential of combining fuzzy clustering and MOEAs on overlapping community detection, and we would like to further explore the proposed EMOFM from the following aspects: Firstly, it is desirable to investigate other distance measures or an ensemble of multiple distance measures to further improve the performance of EMOFM. Secondly, due to the resolution limit in community detection [67], some objectives for measuring the quality of overlapping communities other than Q_{ov} can be adopted. Thirdly, some more efficient search strategies can be developed to reduce the time complexity of EMOFM.

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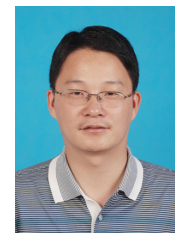
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